DATABASE TECHNOLOGIES – ASSIGNMENT 9

Create appropriate indexes to solve the following problems. Note: use “sales” database to solve below queries.

1. Create an index that will enable a user to pull orders for ‘1990-10-03’ out of the Orders table quickly.

CREATE INDEX date_index ON orders(odate);

SELECT *

FROM orders

WHERE odate = '1990-10-03';

```
[d5_96525_Deven@sales]>CREATE INDEX date_index ON orders(odate);
Query OK, 0 rows affected (1.663 sec)
Records: 0  Duplicates: 0  Warnings: 0

[d5_96525_Deven@sales]>SELECT *
    -> FROM orders
    -> WHERE odate = '1990-10-03';
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3001 |   18.69 | 1990-10-03 | 2008 | 1007 |
| 3003 |  767.19 | 1990-10-03 | 2001 | 1001 |
| 3002 | 1900.10 | 1990-10-03 | 2007 | 1004 |
| 3005 | 5160.45 | 1990-10-03 | 2003 | 1002 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
+------+---------+------------+------+------+
5 rows in set (0.028 sec)

[d5_96525_Deven@sales]>
```

2. If the Orders table has already been created, how can you force the onum field to be unique (assume all current values are unique)?

CREATE UNIQUE INDEX onum_index ON orders(onum);

```
[d5_96525_Deven@sales]>CREATE UNIQUE INDEX onum_index ON orders(onum);
Query OK, 0 rows affected (0.464 sec)
Records: 0  Duplicates: 0  Warnings: 0

[d5_96525_Deven@sales]>
```

3. Create an index that would permit salesperson to retrieve his orders.

SELECT *

FROM orders

WHERE snum = 1001;

CREATE INDEX snum_index ON orders(snum);

SELECT *

FROM orders

WHERE snum = 1001;

```
[d5_96525_Deven@sales]>SELECT *
    -> FROM orders
    -> WHERE snum = 1001;
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3003 |  767.19 | 1990-10-03 | 2001 | 1001 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+------+---------+------------+------+------+
3 rows in set (0.030 sec)

[d5_96525_Deven@sales]>
[d5_96525_Deven@sales]>CREATE INDEX snum_index ON orders(snum);
Query OK, 0 rows affected (0.740 sec)
Records: 0  Duplicates: 0  Warnings: 0

[d5_96525_Deven@sales]>
[d5_96525_Deven@sales]>SELECT *
    -> FROM orders
    -> WHERE snum = 1001;
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3003 |  767.19 | 1990-10-03 | 2001 | 1001 |
| 3008 | 4723.00 | 1990-10-04 | 2006 | 1001 |
| 3011 | 9891.88 | 1990-10-04 | 2006 | 1001 |
+------+---------+------------+------+------+
3 rows in set (0.011 sec)

[d5_96525_Deven@sales]>
```

4. Let us assume that each salesperson is to have only one customer of a given rating, and this is currently the case. Create an index that enforces it.

CREATE UNIQUE INDEX snum_rating_idx

ON customers(cnum, rating);

```
[d5_96525_Deven@sales]>CREATE UNIQUE INDEX snum_rating_idx
    -> ON customers(cnum, rating);
Query OK, 0 rows affected (0.452 sec)
Records: 0  Duplicates: 0  Warnings: 0

[d5_96525_Deven@sales]>
```

5. Create an index to speed up searching order on a given date by given customer.

CREATE INDEX cnum_odate_idx

ON orders(cnum,odate);

```
[d5_96525_Deven@sales]>CREATE INDEX cnum_odate_idx
    -> ON orders(cnum,odate);
Query OK, 0 rows affected (0.733 sec)
Records: 0  Duplicates: 0  Warnings: 0

[d5_96525_Deven@sales]>
```